Demetrius Hernandez

915-245-5751 | dhernan7@nd.edu | demetrius-hernandez.github.io

Professional Summary

Computer science Ph.D. researcher working across artificial intelligence, autonomous systems, and public policy. Combines hands-on technical research with experience in AI safety, governance, and national security to inform the responsible development and deployment of emerging technologies.

Education

University of Notre Dame — Expected Dec. 2028
Ph.D. in Computer Science — *Notre Dame, IN*

University of Texas at El Paso (UTEP) — Dec. 2022
B.S. in Computer Science; Minor in Mathematics; Cum Laude — *El Paso, TX*

Technical Skills

Programming: Python, R, Julia, C++, MATLAB
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn; deep learning, reinforcement learning
Engineering: Git, Linux, Docker, Kubernetes, .NET; requirements engineering, agent evaluation
AI-Assisted Development: Claude Code, Claude Cowork; AI coding tools and agent coordination
Policy: AI governance, human oversight, sociotechnical analysis, UAS regulatory analysis

Work Experience

RAND Corporation — Jun. 2026 – Aug. 2026
Summer Associate, Center on the Geopolitics of AGI — *Washington, DC*

- Built internal AI agents to support robotics and frontier AI analysis, with attention to national security and geopolitical implications.
- Collaborated with senior leadership to develop research directions on advanced robotics and autonomous AI systems.

Notre Dame–IBM Technology Ethics Lab — Aug. 2025 – Present
Research Project Co-Lead, AI-Enabled Drone Governance — *Notre Dame, IN*

- Convened a cross-institutional collaboration with Brown University PI Rich Morales and Notre Dame researchers; co-led the project and co-authored the proposal securing $38,000 for ethical AI-enabled drone deployment.
- Lead sociotechnical research on human–AI collaboration, operational context, and human oversight; lead-author FAA and FCC public comments translating drone research into governance recommendations.

University of Notre Dame — Aug. 2024 – Present
Ph.D. Researcher, Drone Response Lab / SAREC — *Notre Dame, IN*

- Develop reasoning and decision-support approaches for autonomous drones in emergency response, connecting software requirements with safety, ethics, and human oversight.
- Research distributed reasoning agents, stakeholder-driven explainability, and hierarchical metamorphic test oracles for agent evaluation.

White Sands Missile Range — Feb. 2023 – Aug. 2024
Computer Scientist, Counter Drone Team — *White Sands Missile Range, NM*

- Automated counter-drone test analysis; developed graph-theoretic evaluation approaches and a neural-network simulation framework to support testing.
- Secured a $10,000 DoD SMART CREATES grant for custom drone development to strengthen counter-drone testing capabilities.

UTEP Computer Science Department — Jan. 2022 – Dec. 2022
Undergraduate Researcher — *El Paso, TX*

- Built TensorFlow neural-network architectures encoding minimizer schemes to improve DNA assembly and long-read DNA/RNA sequence alignment.

White Sands Missile Range — May 2022 – Aug. 2022
Software Engineering Intern, Survivability and Vulnerability Directorate — *White Sands Missile Range, NM*

- Built instrumentation-control and data-acquisition software for antennas and spectrum analyzers supporting electromagnetic weapons-system testing.

UTEP Brain Computation Lab Sep. 2021 – Jan. 2022
Undergraduate Research Assistant *El Paso, TX*

- Implemented a brain-circuit decision-making model connecting computational analysis with behavioral physiology research.

Oregon State University Jul. 2021 – Sep. 2021
Research Experience for Undergraduates (REU) Researcher *Corvallis, OR*

- Applied machine learning and adapted occupancy models to eBird community-science datasets for quantitative ecology research.

UTEP Cyber-Share Center of Excellence Jul. 2020 – Jul. 2021
Undergraduate Research Assistant *El Paso, TX*

- Used high-performance computing to analyze and visualize the approximately 120-million-atom Cafeteria roenbergensis virus structure.

The Walt Disney Company Aug. 2019 – Jan. 2020
Disney College Program Intern *Lake Buena Vista, FL*

- Combined guest-service responsibilities with engineering exploration seminars and hackathon participation, developing teamwork and technical communication skills.

Selected Publications

Hernandez, D., Ibaraki, K., Morales, R., and Cleland-Huang, J. "Toward Stakeholder-Driven Explainability for Ethically Constrained Physical-AI Decisions." *IEEE Requirements Engineering (RE)*, 2026. Best Paper Honorable Mention.

Hernandez, D., Harris, K., Hernandez, T., and Morales, R. "Navigating the Black Box: Operational Lenses for AI-Enabled Drone Governance." *MIT Science Policy Review* 6, 109–118, 2025. doi:10.38105/spr.3v9f8hi6zq.

Hernandez, D., and Cleland-Huang, J. "Ambient Advisory Models: Augmenting Runtime Models Into Distributed Reasoning Agents." *ACM/IEEE MODELS*, 2025. doi:10.1109/MODELS67397.2025.00028.

Patents

Co-Inventor. "Bayesian-Guided Tactical Reasoning and Adaptive UAV Swarming for Clue-Based Search and Rescue." U.S. Patent Application No. 19/689,701, filed May 27, 2026. Patent pending.

Selected Policy Contributions

Lead Author. Formal public comment submitted to the FCC in response to "Unleashing American Drone Dominance," Public Notice DA 26-314, GN Docket No. 26-74 (2026).

Lead Author. Drone Response Lab public comment submitted to the FAA in response to "Normalizing Unmanned Aircraft Systems Beyond Visual Line of Sight Operations," Notice No. 25-07, Docket No. FAA-2025-1908 (2025).

Awards and Fellowships

National Recognition: DoD SMART Scholar of the Year (2024); Certificate of Special Congressional Recognition (2024); White Sands Test Center Commander's Coin for Excellence (2024)

Research Awards: IEEE RE Best Paper Honorable Mention (2026); ICSE Student Research Competition, Second Place (2026)

Other Selected Awards: U.S. Army TRADOC Mad Scientist Silver Award (2026); RISE AI Best Poster (2025); Notre Dame Graduate Research Symposium Best Oral Presentation (2025)

Fellowships: NSF CSGrad4US (2024; three years of support); Sorin Fellow (2024); DoD SMART Scholarship (2021)

Leadership and Service

Graduate Student Representative, Notre Dame Engineering College Council | 2026–Present
Network Member, Irregular Warfare Center Emerging Tech Functional Area Network | 2026–Present
Communications Director, Notre Dame Science Policy Initiative | 2025–2026
Graduate Student Board Member, Notre Dame Computer Science Department | 2025–Present
Inclusive Excellence and Engagement Committee Member, Notre Dame Engineering | 2025–Present
Advisory Board Member, UTEP Center for Research in Engineering and Technology | 2024–Present